

Regular routes for total lengths 2 to 6

Primes have one representation type; composites have more.

N = 2 | Prime

1 representation type



$$(k, s) = (2, 1)$$

Unit steps

N = 3 | Prime

1 representation type



$$(k, s) = (3, 1)$$

Unit steps

N = 4 | Composite

2 representation types



$$(k, s) = (2, 2)$$

Non-unit steps

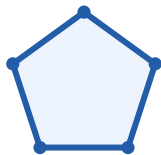


$$(k, s) = (4, 1)$$

Unit steps

N = 5 | Prime

1 representation type



$$(k, s) = (5, 1)$$

Unit steps

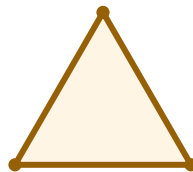
N = 6 | Composite

3 representation types



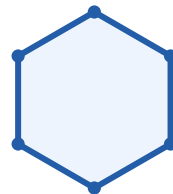
$$(k, s) = (2, 3)$$

Non-unit steps



$$(k, s) = (3, 2)$$

Non-unit steps



$$(k, s) = (6, 1)$$

Unit steps

Each (k, s) records k steps of length s . All routes use the same length scale.

The digon traverses one segment twice; its arrows lie on the coincident edges.